3.20 Local Rings

Local rings arise quite naturally when localizing at a prime ideal (see (3.6.42) and (3.20.5)) so we recall some basic properties here.

Definition 3.20.1 (Local Ring)
*A ring A is a **local ring** if it has a unique maximal ideal $\mathfrak{m}$. The field $A/\mathfrak{m}$ is called the **residue field** of A and sometimes denoted $\kappa(\mathfrak{m})$.*

When A is a k-algebra we may denote the residue field by $k(\mathfrak{m})$.

Definition 3.20.2 (Local Homomorphism)
*Let $(A, \mathfrak{m}_A)$ and $(B, \mathfrak{m}_B)$ be local rings. A ring homomorphism $\phi : A \to B$ is said to be a **local homomorphism** if*

$$\phi(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$$

Recall that the group of units $A^\star$ of a ring is a saturated multiplicative set, that is

$$xy \in A^\star \iff x \in A^\star \wedge y \in A^\star$$

Proposition 3.20.3 (Criteria for Local Rings)
Let A be a ring. Then the following are equivalent

a) A is a local ring
b) $A \setminus A^\star$ is an additive subgroup of A

In this case $\mathfrak{m} = A \setminus A^\star$ is the unique maximal ideal of A.

Proof. a) $\implies$ b) Let $\mathfrak{m}$ be the unique maximal ideal then, because it's proper, $\mathfrak{m} \cap A^\star = \varnothing \implies \mathfrak{m} \subseteq A \setminus A^\star$ by (3.3.13). Conversely given $x \in A \setminus A^\star$ then (x) is a proper ideal by (3.3.32), and therefore contained in a maximal ideal (3.3.15) which by uniqueness means $x \in \mathfrak{m}$.

b) $\implies$ a) Define $\mathfrak{m} = A \setminus A^\star$ it's a (prime) ideal because it is an additive subgroup and $A^\star$ is a saturated multiplicative set. Let $\mathfrak{a}$ be a proper ideal then $\mathfrak{a} \cap A^\star = \varnothing \implies \mathfrak{a} \subseteq \mathfrak{m}$. Therefore $\mathfrak{m}$ is the unique maximal ideal. ∎

Proposition 3.20.4 (Quotient of a Local Ring is Local)
Let $(A, \mathfrak{m})$ be a local ring and $\mathfrak{a} \lhd A$ a proper ideal. Then $A/\mathfrak{a}$ is a local ring, with maximal ideal $\mathfrak{m}/\mathfrak{a}$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

As $\mathfrak{a}$ is proper, it is contained in some maximal ideal (3.3.15), which must be $\mathfrak{m}$ by uniqueness; so $\mathfrak{a} \subseteq \mathfrak{m}$, and by the ideal correspondence of (3.3.55), the maximal ideals of $A/\mathfrak{a}$ correspond to the maximal ideals of A containing $\mathfrak{a}$ – of which $\mathfrak{m}$ is the only one (as A is local) – so $A/\mathfrak{a}$ has the unique maximal ideal $\mathfrak{m}/\mathfrak{a}$, that is, is local. ∎

Example 3.20.5
Let A be a ring and $\mathfrak{p} \lhd A$ a prime ideal. Then $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

When $A \subset K$ is a subring of a field then $A \subset A_\mathfrak{p} \subset K$ in a natural way.

We may use this to provide another criteria

Lemma 3.20.6 (Criteria for Local Domain)
Let $A \subset K$ be a subring of a field with a prime ideal $\mathfrak{p} \lhd A$. Then $A \subset A_\mathfrak{p}$.

A is a local ring with unique maximal ideal $\mathfrak{p}$ if and only if $A = A_\mathfrak{p}$,

Proof. We've observed that $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

If $A = A_\mathfrak{p}$ then it is a local ring and $\mathfrak{p} \subset \mathfrak{p}A_\mathfrak{p}$. For $y \notin \mathfrak{p}$, then $\frac{1}{y} \in A_\mathfrak{p} \implies \frac{1}{y} \in A$. So we see that $x \in \mathfrak{p}, y \notin \mathfrak{p}$ we have $\frac{x}{y} \in \mathfrak{p}$ and $\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$.

Conversely suppose A is a local ring with unique maximal ideal $\mathfrak{p}$. Then $y \notin \mathfrak{p} \implies y \in A^\star$ and $A_\mathfrak{p} = A$ as required. ∎